PARTNERSHIP UPDATE

Milestone Update

What we built. Why. How. What's next.

For the peaq tech, DePIN, and partnership team

AXI MOBILITY LIMITED

A fleet platform built on peaq

Nairobi, Kenya

Prepared for:	peaq tech / DePIN / partnership team
Pilot context:	ROAM Electric — 10 internal EV motorcycles, Nairobi
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Contact:	team@aximobility.com

Apache-2.0 open source · Repos: `peaq-contracts` + `peaq-integration`

1 Summary

AXI runs a fleet platform. We monitor vehicles, drivers, batteries, and route data, then turn that into alerts, actions, and reports for the fleet owner.

1.1 What we built on peaq

Three smart contracts on Agung testnet:

- **Insurance Risk Oracle** - publishes a per-driver risk score that insurers can read.
- **High-Risk Command Vault** - requires multiple sign-offs before a fleet-wide action runs.
- **Credential Revocation Registry** - lets issuers cancel a license or certificate so anyone can check it instantly.

Plus a TypeScript library and command-line tool that talk to peaq's identity, role, and storage features, sign with hardware-backed keys, and post a daily summary of all events to the chain.

1.2 Where we are today

Status

- **51 tests passing** on the local test setup.
- **Agung dry-run deploy successful.** 25 minutes from a blank config to the first signed event.
- **About 80% of the pre-audit checklist is closed.** Remaining: extra security testing, multi-signature wallet setup, external auditor, and a frozen review tag.
- **Going live on mainnet** needs a 4-of-7 multi-signature wallet and an external audit. Realistic window: 6 to 10 weeks from audit start.

2 Why peaq

We picked peaq because it ships features built for mobility out of the box. Other chains would force us to build (and maintain) those features ourselves.

2.1 The decision matrix

What we needed	What peaq gives	Cost on a generic chain
A trusted identity for every vehicle and driver	`peaqDid` (built-in identity system)	Build our own identity registry
Roles for operators, regulators, and auditors	`peaqRbac` (built-in role system)	Write and maintain a permissions contract
A way to record evidence so anyone can verify it	`peaqStorage` + smart contracts on the same chain	Pick one tool or run two chains
Realistic gas and finality for African DePIN	Substrate finality + EVM precompiles in one node	Bridge fees and two-chain ops
Mobility ecosystem alignment	DePIN focus, ROAM-class peers	Generic finance and DeFi peers

Strategic lock

We are not running on two chains. **peaq is the main chain.** We post a daily summary to Polygon as a backup only - in case peaq has an outage. This is a deliberate, long-term choice.

3 What we built (three use cases)

Each contract solves a real problem from our ROAM pilot.

3.1 Insurance Risk Oracle

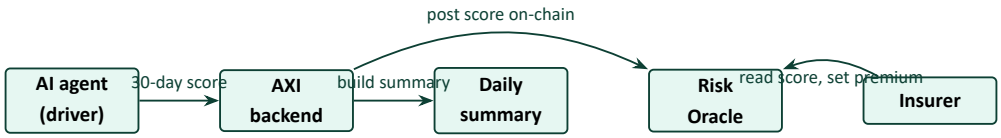
3.1.1 The problem

Insurers in Kenya price premiums on patchy fleet data, so they add a 15-25% surcharge they call "uncertainty risk". ROAM and every fleet we spoke to overpays.

3.1.2 The solution

We post a risk score (0-1000) for each driver to the chain. Insurers read the latest score before quoting. The log is append-only and signed - they can audit every score themselves.

3.1.3 How it works



3.1.4 Example

ROAM rider Mary Akinyi

6 months riding · 4,200 km · 0 harsh-brake events · 89% safety score.

AXI posts: score 920, expires in 30 days.

Insurer reads the score → Mary qualifies for the lowest premium tier.

3.1.5 Why on-chain, not in our database

Insurers will not trust AXI's database - we are an interested party. They will trust a public log they can read and audit independently.

3.1.6 Roles

Contract	`InsuranceRiskOracle.sol` (upgradeable, append-only)
`ATTESTOR_ROLE`	AXI's service that publishes scores
`CALIBRATOR_ROLE`	Risk team that tunes the score-to-premium curve
`UPGRADER_ROLE`	Multi-sig wallet (4-of-7) that ships upgrades
`DEFAULT_ADMIN_ROLE`	Multi-sig wallet that grants and revokes roles
Score range	0 to 1000
Freshness rule	Score must be less than 7 days old, otherwise rejected

3.2 High-Risk Command Vault

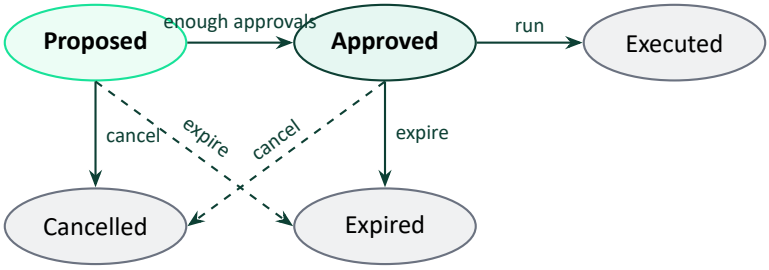
3.2.1 The problem

Fleet-wide actions are dangerous. ”Lock 47 bikes” is one click, and it has to be reversible, auditable, and need more than one person to run. Most operator dashboards have a single-password override - that is how fleets get held ransom.

3.2.2 The solution

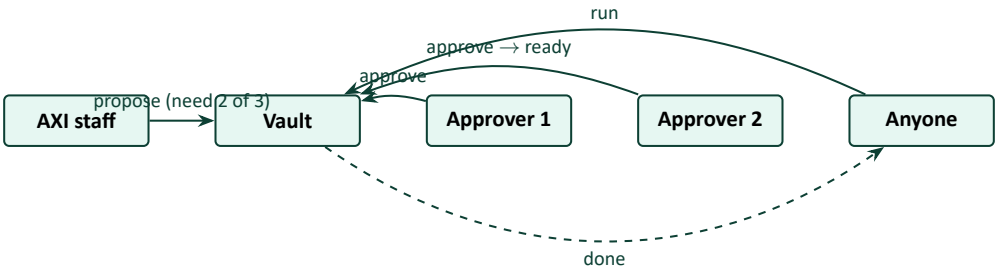
A multi-person approval gate. AXI staff propose a high-risk command (lock the fleet, override every route, send a mass broadcast, withdraw funds). A pre-set group of approvers must sign. Once enough approve (e.g. 2 of 3), anyone can run the command - including a regulator if AXI staff disappear.

3.2.3 States the command goes through



Bright green = initial · Gray = terminal · Dashed = also reachable

3.2.4 How it works



3.2.5 Example

Mary's bike battery overheats

KMG 117P shows battery temperature 52°C - fire risk. Brian proposes "lock the 3 bikes in this depot" (needs 2 of 3 approvals).

- Joy approves · 1 of 2 in
- ROAM ops manager approves · 2 of 2 → ready to run
- Anyone runs the command → bikes locked

The chain shows who proposed, who approved, when, and what.

3.2.6 Why on-chain, not Slack approvals

A Slack approval can be denied later. An on-chain approval cannot. The audit trail is the product - insurers and regulators want proof of good governance before they price premiums or grant licences.

3.2.7 Roles

Contract	`HighRiskCommandVault.sol` (**not upgradeable**)
Approval threshold	1 to 10 approvers (set per command)
Expiry window	1 minute to 30 days
Replay safety	Each command has a unique ID, so the same command cannot run twice
Payload check	The proposal locks the exact payload; running it must use the same bytes
Anyone can run it	Once approved, any address can run it - no AXI lock-in

Why not upgradeable

An upgradeable approval gate is a back door. The Vault is permanent - operators see the exact rules they are approving, and AXI cannot silently change them later.

3.3 Credential Revocation Registry

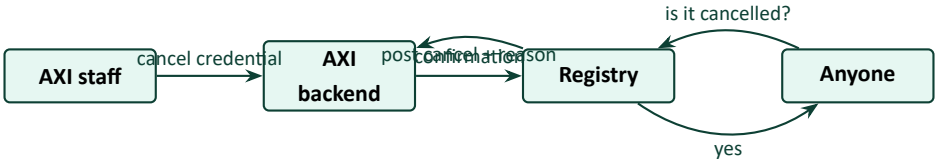
3.3.1 The problem

Driver licences, NTSA inspection certificates, and insurance binders expire or get revoked. Today fleets find out at the roadside check (around $300 fine + tow-truck bill). Insurers find out at claim time (around $5,000 loss). Verification is still paper-based.

3.3.2 The solution

Each credential gets a unique fingerprint stored on-chain. When the issuer (NTSA, insurance, fleet manager) cancels it, they post the cancellation. Anyone - insurer, regulator, the rider app - can instantly check whether a credential is still valid. Only the original issuer can cancel their own credential.

3.3.3 How it works



3.3.4 Example

ROAM rider Wanjiku's NTSA inspection

Inspection expires on 2026-05-15. The NTSA officer cancels it on-chain (reason: expired).

At the next checkpoint, the inspector scans the QR code on Wanjiku's bike. The app checks the registry → cancelled. Wanjiku gets a polite warning, not a fraud charge. AXI avoids a court date.

3.3.5 Why on-chain, not the NTSA portal

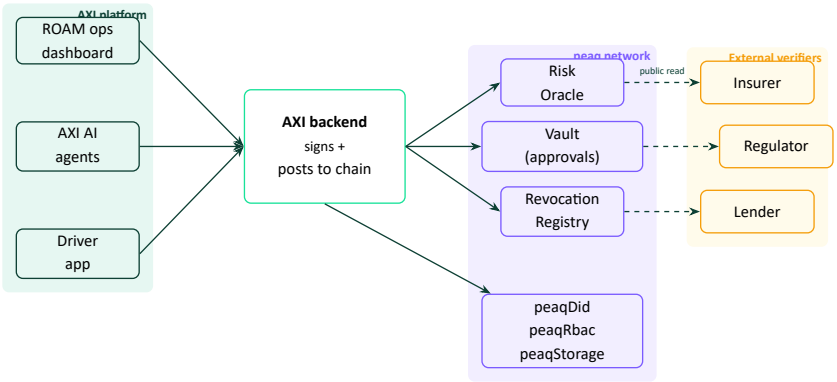
NTSA's portal goes down often. The chain does not. And insurers outside Kenya can verify without trusting any single Kenyan database.

3.3.6 Roles

Contract	`CredentialRevocationRegistry.sol` (upgradeable)
`ISSUER_ROLE`	Only the original issuer can cancel a credential they issued
`revoke()`	Cannot cancel the same credential twice (preserves the first reason)
`revokeBatch()`	Cancel up to 100 at once (useful when rotating an issuer's key)
`unrevoke()`	Admin-only emergency undo (used if an issuer key is compromised)
`pause()`	Emergency stop

4 How it all fits together

4.1 The architecture



4.2 Three flows, every day

- **Daily summary at midnight UTC** — AXI builds one fingerprint of the day's events and posts it to peaq (and to Polygon as backup). Anyone can pick a random event from yesterday and prove it was in that summary.
- **Per-event signing (live)** — when an AI agent flags a serious event, the AXI backend signs it with a hardware-backed key and posts the proof to peaq within seconds. The case UI shows a "verified" badge.
- **Public verification (anytime)** — `verify.aximobility.com/api/verify/[hash]` returns proof anyone can check against peaq directly. No trust in AXI needed.

5 The peaq footprint we use

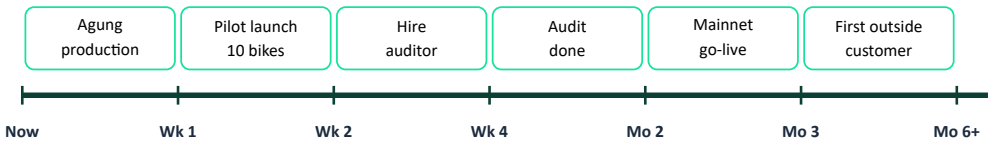
peaq feature	How AXI uses it	Frequency
Smart contracts	The 3 contracts above (Oracle, Vault, Registry)	1-10x/day
`peaqDid`	A unique identity per vehicle, driver, and asset	~50/day in pilot
`peaqRbac`	Roles for issuers, signers, approvers, auditors	per role change
`peaqStorage`	Storing compliance evidence we can re-fetch later	monthly
EVM compatibility	Standard Solidity tooling works out of the box	every call
Mainnet RPC	Daily summary + live event signing	continuous
Agung testnet	Where we test before shipping to mainnet	continuous

6 Milestones reached

Date	Milestone	Status
2026-02	Locked peaq as our main chain (Polygon as backup)	✓
2026-02	Verified 34 facts about peaq, flagged 7 open questions for the team	✓
2026-02	Initial architecture and integration design	✓
2026-03	Built the TypeScript library: identity, roles, storage, signing, retries, logging	✓
2026-03	Built a command-line tool with 5 command groups (anchor, did, storage, health, status)	✓
2026-03	Wrote 3 production-shape smart contracts with role-based access and clear error messages	✓
2026-04	51 tests passing on the local test setup	✓
2026-04	Security scanner running in CI on every change	✓
2026-04	Agung dry-run deploy successful	✓
2026-04	Pre-audit checklist about 80% complete	✓
2026-05	Step-by-step deploy guide (25 minutes from zero to first signed event)	✓
2026-05	Per-contract security analysis published	✓
2026-05	Responsible disclosure policy published	✓
2026-05	Technical review packet for the peaq tech team	✓
2026-05	This milestone document	✓

7 Roadmap

7.1 Visual timeline



7.2 Roadmap detail

Milestone	When	What it unlocks
Production Agung deploy	Week 1	Live testing on real Agung; pilot can sign daily
Internal pilot launch	Week 2	First daily summary, first risk score, first cancellation
Hire external auditor	Week 3-4	Freeze contracts for audit
Audit + fix findings	Week 4-8	Last gate before mainnet
Go live on peaq mainnet	Month 2	Daily summary live, multi-sig wallet in charge
First outside customer	Month 3	Same platform, different fleet type
Bug bounty open	Month 3	Public security testing
First customer on dedicated infra	Month 6+	Their own AI hardware

8 What we need from the peaq team

8.1 Answers to seven open questions

Cost + capacity

1. **Exact transaction fees.** What does it cost to add an identity attribute or store an item, in PEAQ tokens and USD?
2. **Bulk-write pricing.** Is there a discount tier for high-frequency writes? At full scale (about 10,000 vehicles) we will write 10-100 events per second.

Identity standards

3. **Standard library compatibility.** Do peaq identities work inside the standard W3C verifiable-credential libraries that insurers already use? This is critical for them to consume our identities directly.

Network safety

4. **Decentralisation numbers.** Validator count, where they are located, how concentrated voting power is.
5. **Major incidents since launch.** Forks or outages we should know about for our risk model.
6. **KREST to PEAQ migration plan.** What is the guidance for partners building on the network now?

Tooling

7. **Official chain definition for viem.** Are you planning to ship one? We are building our own today.

8.2 Code review

- Check our identity issuer and resolver against your spec.
- Check our pallet wrapper code (storage and roles).
- Spot-check our smart contracts for anything you have seen go wrong elsewhere.

8.3 Help with going to mainnet

- Which audit firms do you recommend? (We are considering Trail of Bits, OpenZeppelin, Halborn, Sigma Prime.)
- What monitoring should we have in place before going live?
- Multi-sig wallet setup recommendations (signers, threshold).
- Does peaq announce partner mainnet launches publicly?

8.4 Joint marketing

- Joint announcement when we go live with ROAM (blog post + social).
- Listing on peaq's partners page.
- Reference write-up: we are happy to publish "how AXI uses peaq" as a guide for other partners.

Time-sensitive

The ROAM pilot launches in 2 weeks. We want our daily summary running on Agung before then. A 1-hour call this week unblocks the timeline. Email us at team@aximobility.com.

9 Next actions

9.1 For the peaq team

1. Read this document (15-minute scan).
2. If you want more technical depth, read `PEAQ_REVIEW.md` in the repo (about 90 minutes).
3. Reply to team@aximobility.com with availability for a 1-hour call this week.
4. Send written answers to the 7 open questions, or come ready to discuss them on the call.

9.2 For AXI (in parallel)

- Deploy to production Agung with a funded wallet and multi-sig admin.
- Close the last 20% of the pre-audit checklist (extra automated tests, more coverage).
- Hire an external audit firm (we are picking from Trail of Bits, OpenZeppelin, Halborn, Sigma Prime, pending peaq's recommendation).
- Freeze the contracts and tag the commit before the audit starts.

9.3 Repository links

peaq-contracts	github.com/aximobility/peaq-contracts
peaq-integration	github.com/aximobility/peaq-integration
Frozen tag	`peaq-review-2026-05-07`
Licence	Apache-2.0 (both repos)

9.4 Documents in this packet

File	Purpose
This document	Milestone update for the peaq team (you are reading it)
`MILESTONES.md`	Same content, in plain Markdown for GitHub
`PEAQ_REVIEW.md`	The technical deep-dive (13 sections)
`AUDIT_CHECKLIST.md`	Pre-audit checklist (about 80% complete)
`AGUNG_DEPLOY.md`	Step-by-step deploy guide (25 minutes start to finish)
`SECURITY.md`	How to report a security issue
`threat-models/`	Per-contract security analysis
`VERIFIED_SPECS.md`	34 verified facts about peaq + the 7 open questions

10 Contributors

Name	Role
Brian Mwai	Founder & CEO
Rhyl Kiio	CTO
Raffa	Founding Engineer
Bleyle	Blockchain Engineer (peaq grant primary technical contact)
Mercy Akwera	Sales Lead

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